

Seneca College

School of Information Technology Administration & Security - SITAS

CSN405 - Wireless Networks

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Lab 2: Demonstration of Absorption, Multipath, and Phase

Objectives: The main objective of this lab is to help you to be familiar with the following:

1. Attenuation effect of materials due to absorption
2. Effect on received signal amplitude due to various phases of two signals arriving at the same time
3. RF signal propagation behaviors

After completing this lab, you will be able to:

1. Discuss the attenuation effect of materials due to absorption
2. Evaluate the effect of phase difference and multipath
3. Identify different signal propagation behaviors

Materials Required:

1. A computer with Internet access
2. Simulation software: EMANIM Tool (provided)

Notes:

1. This is individual work. You need to finish this lab and submit your work on blackboard by the due date.
2. Edit this lab WORD document and answer the questions using **bold blue fonts**. Insert the required screen shots in the same document as well.
3. Submissions by email will not be considered. Zero mark will be granted for submissions after due date.
4. After finishing your lab, write your name on the document as follows: **Your First Name. Your Last Name - Lab 2**. Drop your lab in the corresponding drop box.

Part 1: In this part, you will use a simulation tool called EMANIM to view the attenuation effect of materials due to absorption. EMANIM is a free program found on the Course Textbook's. It is posted on the Black Board/ Course Website.

Exercise 1: Visual Demonstration of Absorption

Perform the following steps:

1. Download and install the EMANIM program by double-clicking emanim_setup.exe.
2. From the main EMANIM menu, click Phenomenon

3. Click Sybex CWNA Study Guide

4. Click Exercise E

When a radio wave crosses matter, the matter absorbs part of the wave. As a result, the amplitude of the wave decreases. The **extinction coefficient** determines how much of the wave is absorbed by unit length of material.

5. Vary the length of the material and the extinction coefficient for Wave 1 to see how it affects the absorption. Answer the following questions:

A. How does material length affect the absorption rate?

More RF signal absorption happens when a signal passes through an object with a longer material length than it does via an object with a shorter material length.

B. How does the extinction factor affect the absorption rate?

The wave's absorption capacity per unit length of material is determined by the extinction coefficient. The statement implies that an object with a higher extinction coefficient will absorb more energy than one with a lower coefficient.

C. Define the extinction coefficient.

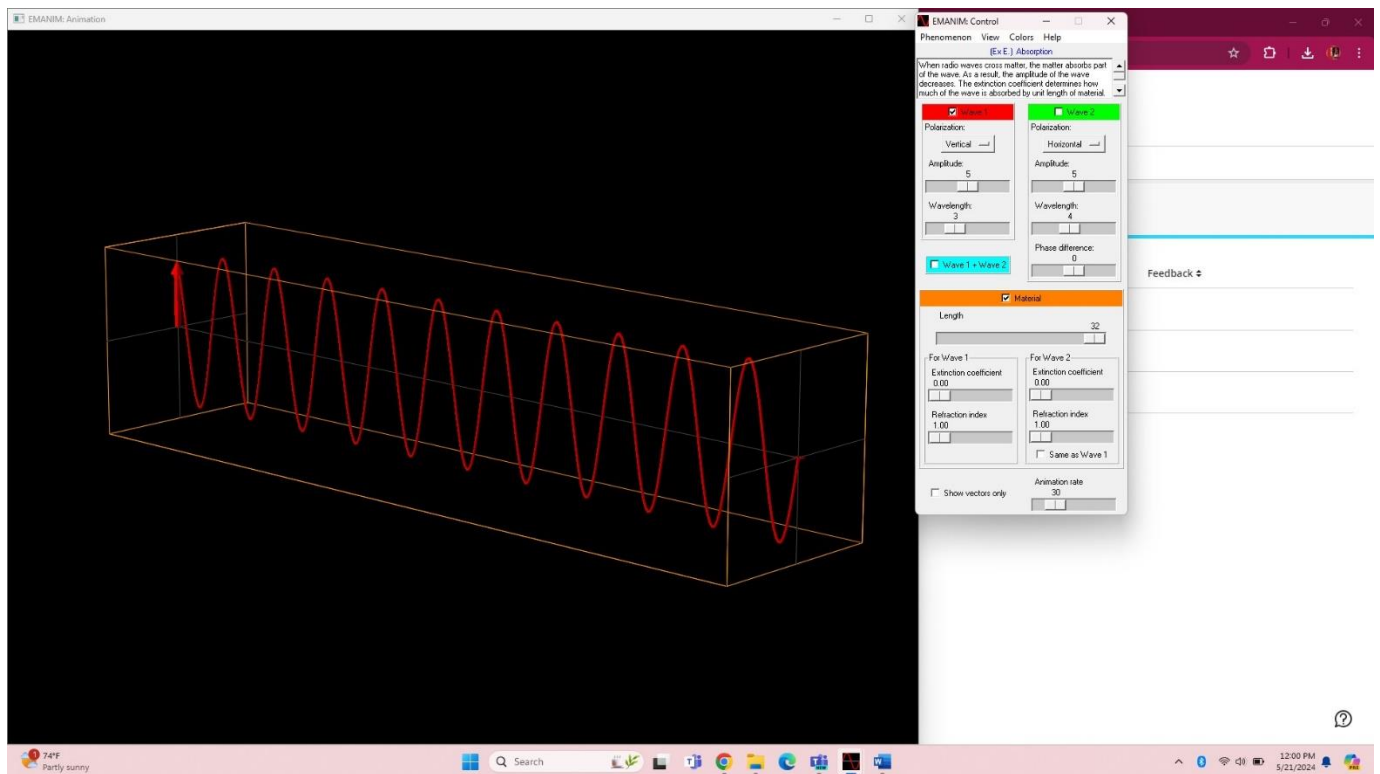
The amount of energy absorbed by a material per unit length when a wave travels through it is known as its extinction coefficient.

Provide screen shots for the following cases, and **provide your comments** on each one of them:

1. Material length = 32 and Extinction Coefficient = 0

The wave's amplitude did not change. It indicates that absorption occurred because wave went through the substance as seen in the picture below

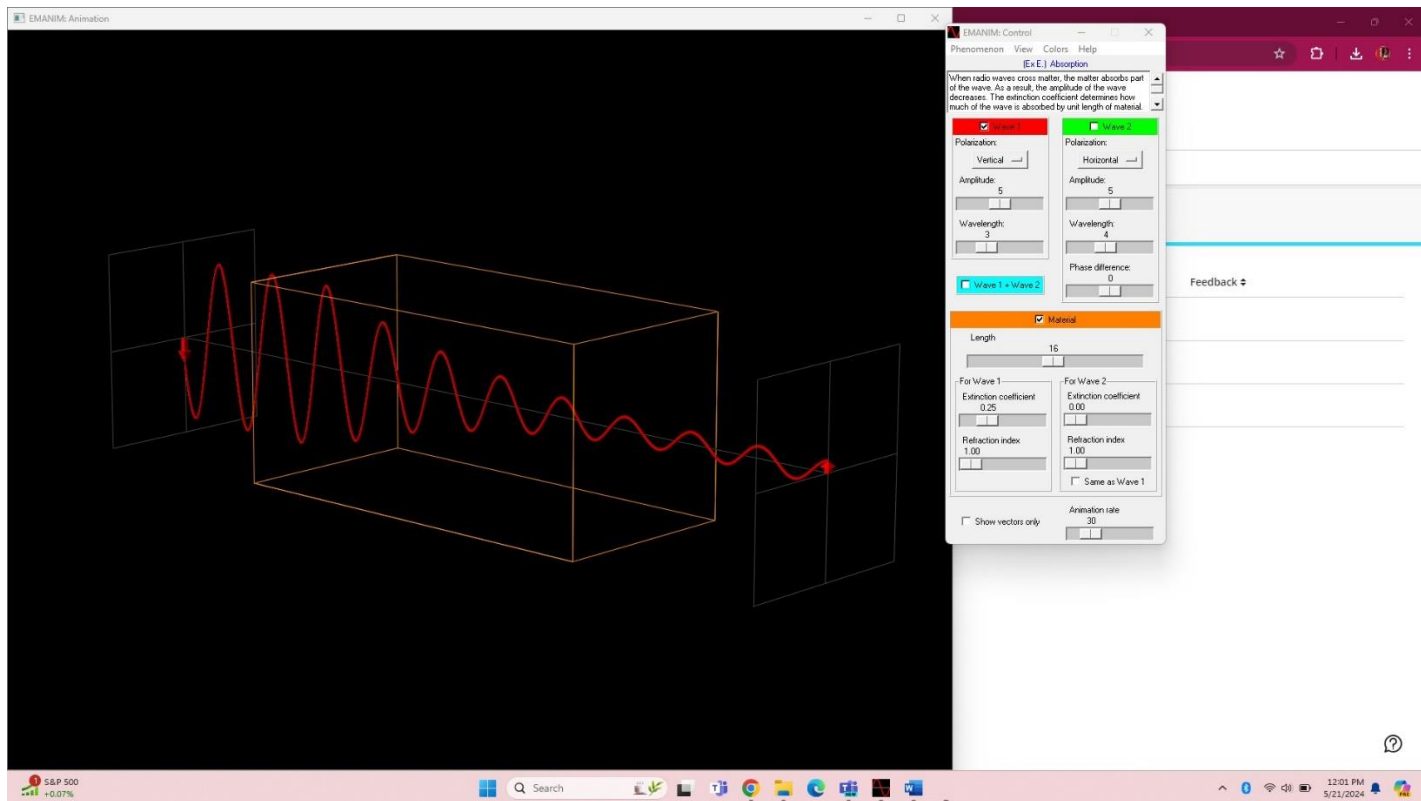
Insert Screen Shot:



2. Material length = 16 and Extinction Coefficient = 0.25

The RF signal was somewhat absorbed when it passed through the material, as indicated by the wave's decreased amplitude, as seen in the picture below.

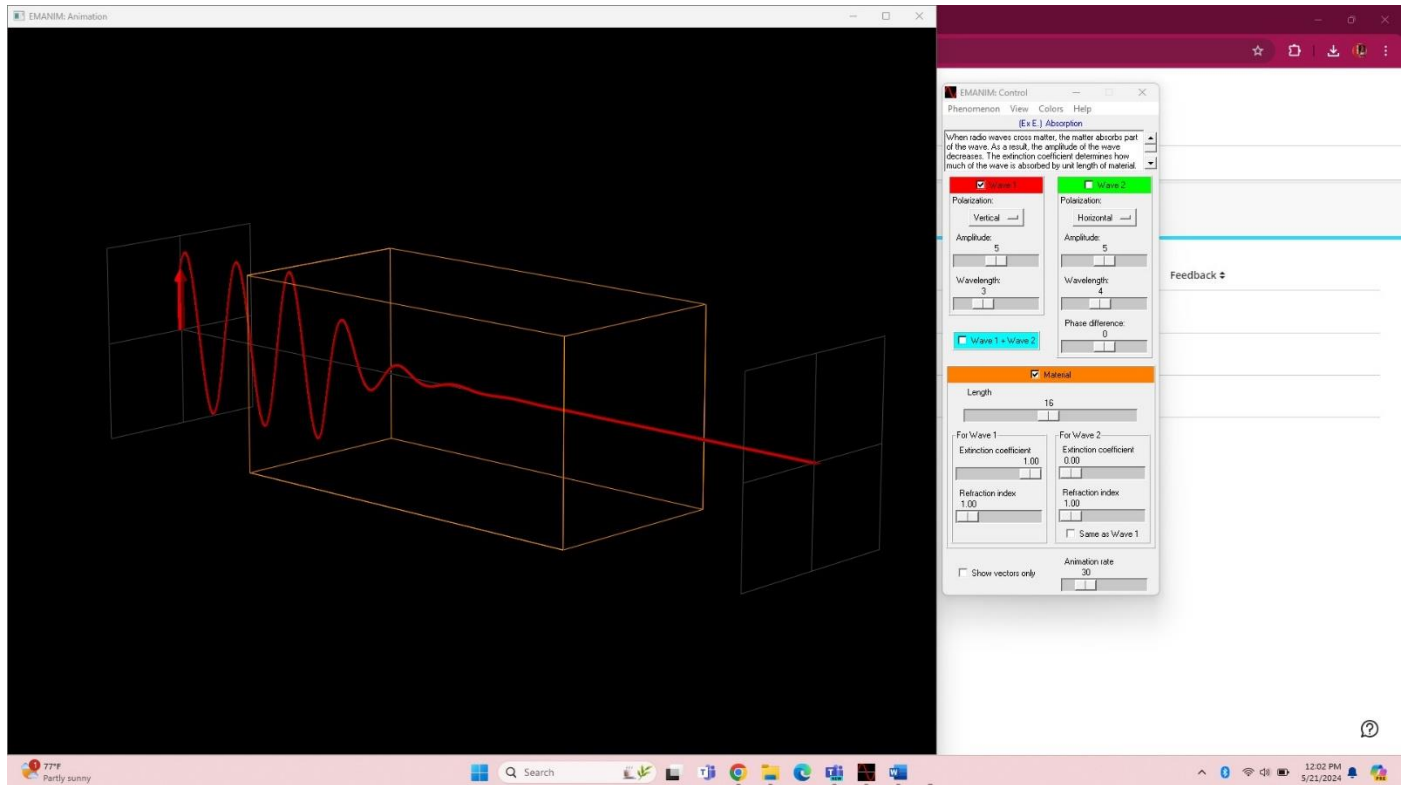
Insert Screen Shot:



- Material length = 16 and Extinction Coefficient = 1

A greater amount of signal absorption occurred nearly 100% of the signal . We can see this in the picture below.

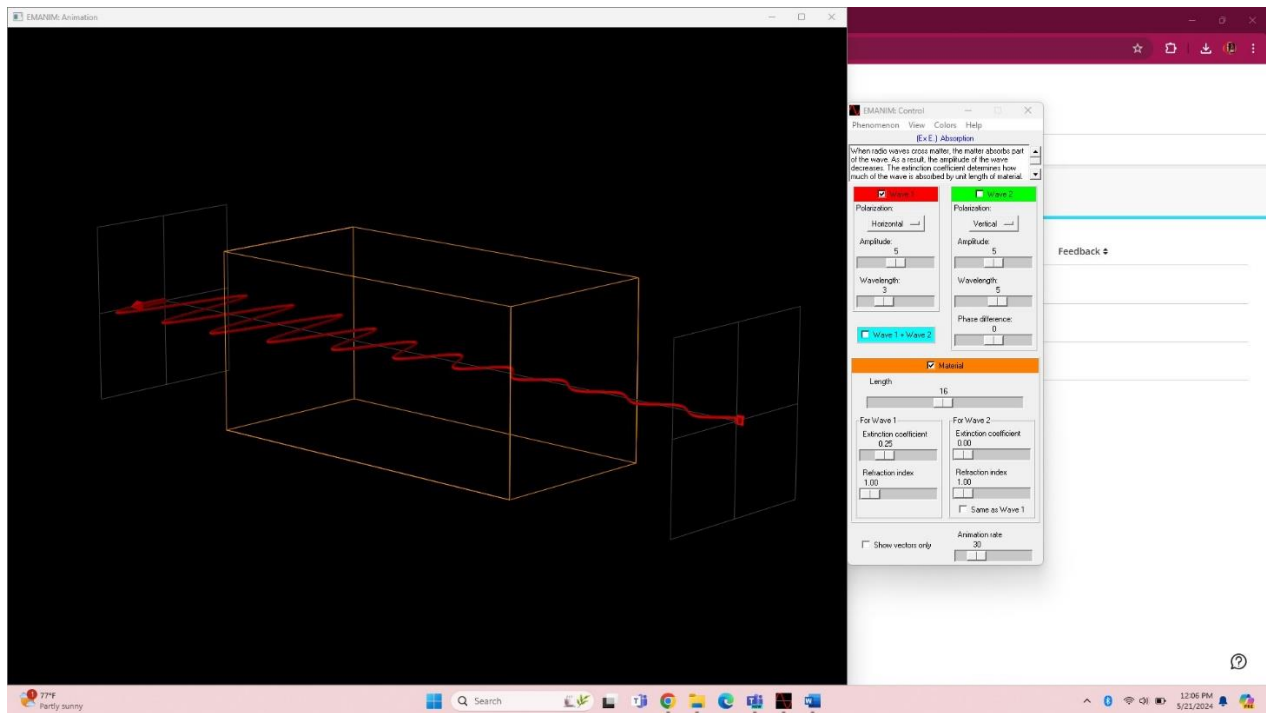
Insert Screen Shot:



4. Keep the Material length = 16, Extinction Coefficient = 0.25, and change the signal polarization to vertical, horizontal, right circular, and left circular. Comment on your observations and provide one screen shot for horizontal polarization only.

Although the absorption rate was consistent across all scenarios, the wave's behavior varied when the material length and extinction coefficient were maintained constant and only the signal polarization was altered. The wave's electric field vector oscillated vertically once the polarization signal was maintained at that level. The vector representing the electric field oscillated horizontally while it was 0. The direction of rotation of the electric field was clockwise in the right circular and counterclockwise in the left.

Insert Screen Shot:



Question: What is meant by signal polarization?

The direction in which the electric field vector of the signal's electromagnetic waves oscillates is known as signal polarization.

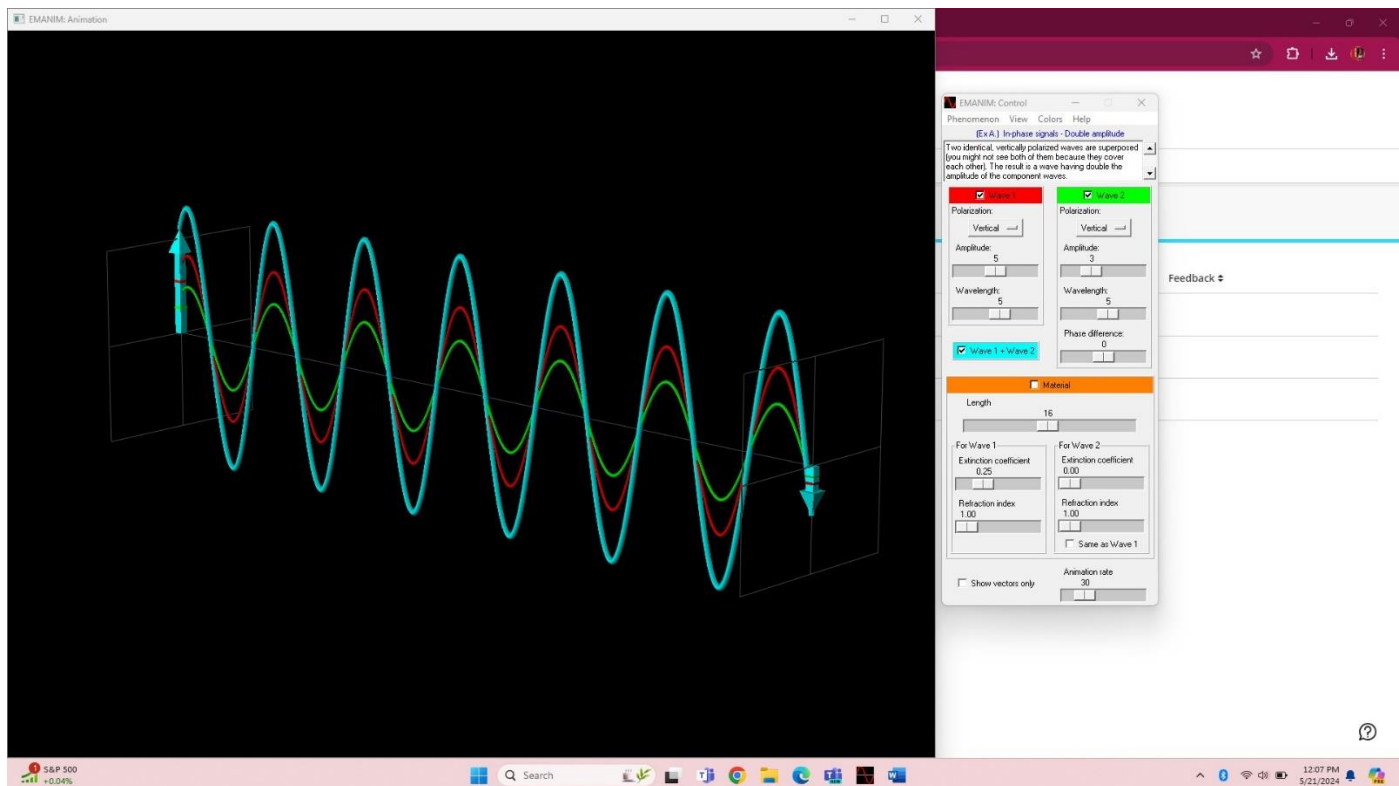
Exercise 2: Visual Demonstration of Multipath and Phase

Use the same tool to view the effect on amplitude due to various phases of two signals arriving at the same time. Perform the following steps:

1. From the main EMANIM menu, click Phenomenon.
2. Click Sybex CWNA Study Guide.
3. Click Exercise A.

Two identical, vertically polarized waves are superposed (you might not see both of them because they cover each other). The result is a wave having double the amplitude of the component waves. Change the amplitude of Wave 2 to 3 and take a screen shot.

Insert Screen Shot:



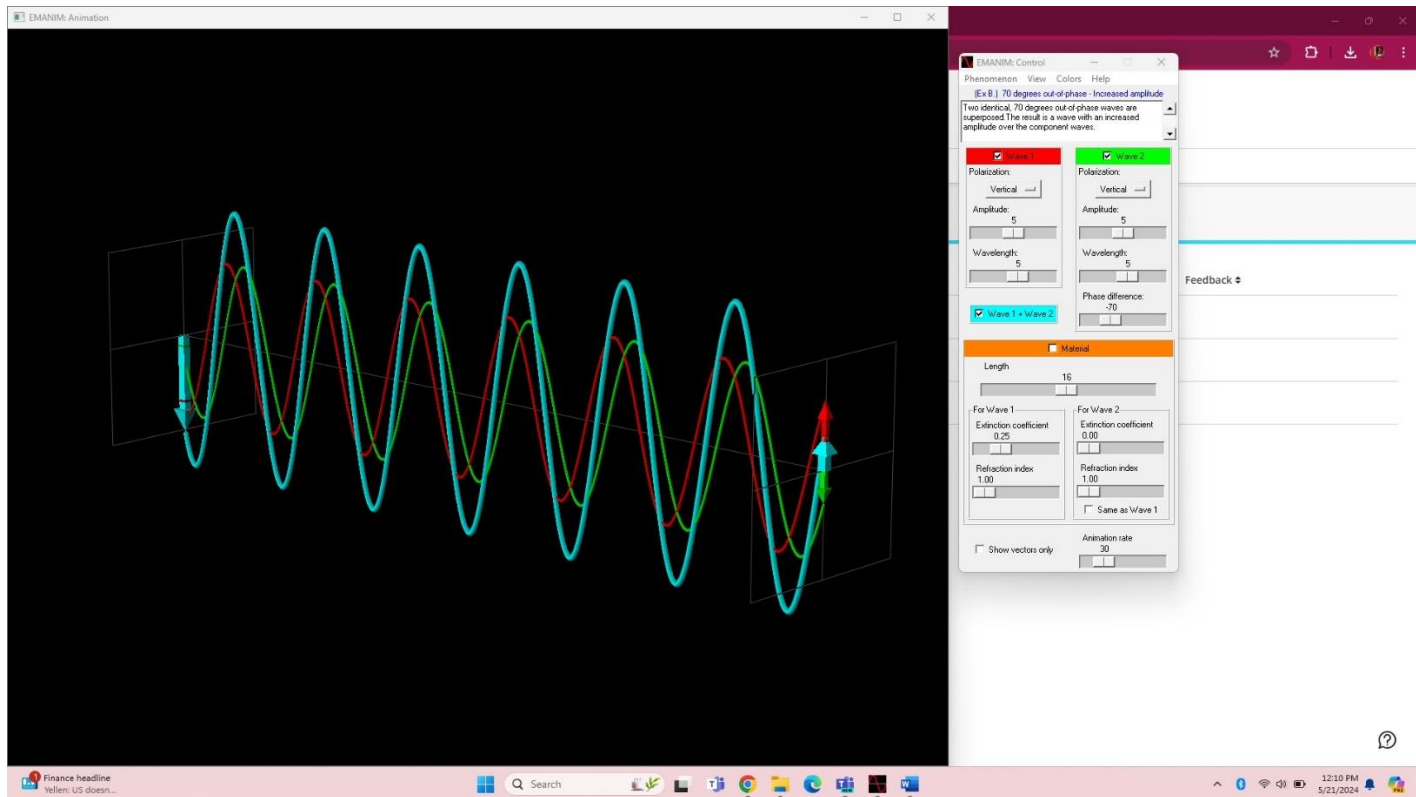
Question: What do we call such case, when we have two signals enhancing each other? Explain your answer and provide values of the phase difference that yield enhancement.

Such a situation is known as **constructive multipath**. As seen in the image, there is an increase in signal intensity when multiple RF signal routes arrive to the receiver simultaneously, either partially or fully out of phase. The resulting wave, represented by the blue line, is stronger than the two individual waves. When there are 0–120-degree phase discrepancies between waves, it happens.

4. Click Exercise B.

Two identical, 70-degree out-of-phase waves are superposed. The result is a wave with an increased amplitude over the component waves.

Insert Screen Shot:



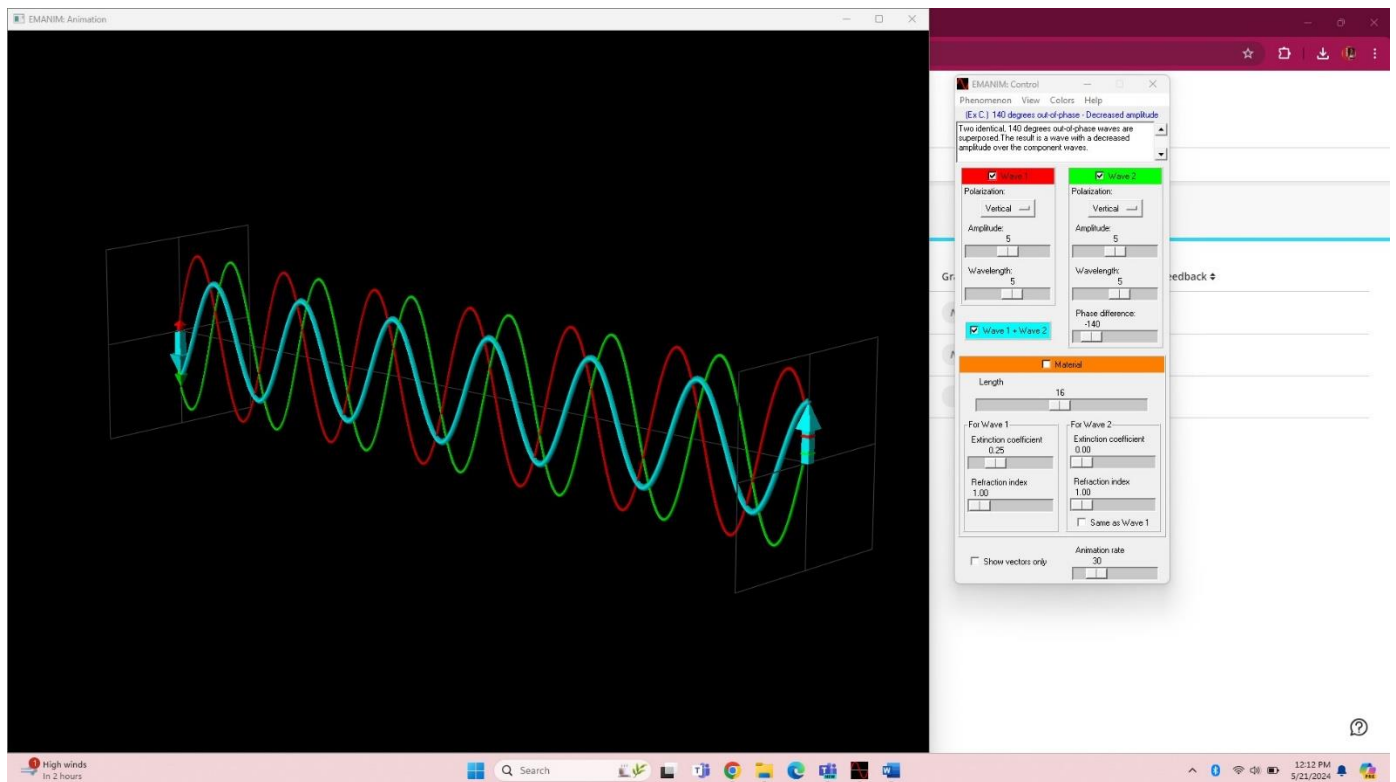
Question: Is there any difference in the resultant waveform if you change the phase difference from -70° to $+70^\circ$? Explain your answer.

The waveform that results from changing the phase difference from -70 to $+70$ is the same. The outcome of the phase shift between the two waves will remain constructive multipath even if the phase shift direction changes from -70 to $+70$.

5. Click Exercise C.

Two identical, 140-degree out-of-phase waves are superposed. The result is a wave with a decreased amplitude over the component waves.

Insert Screen Shot:



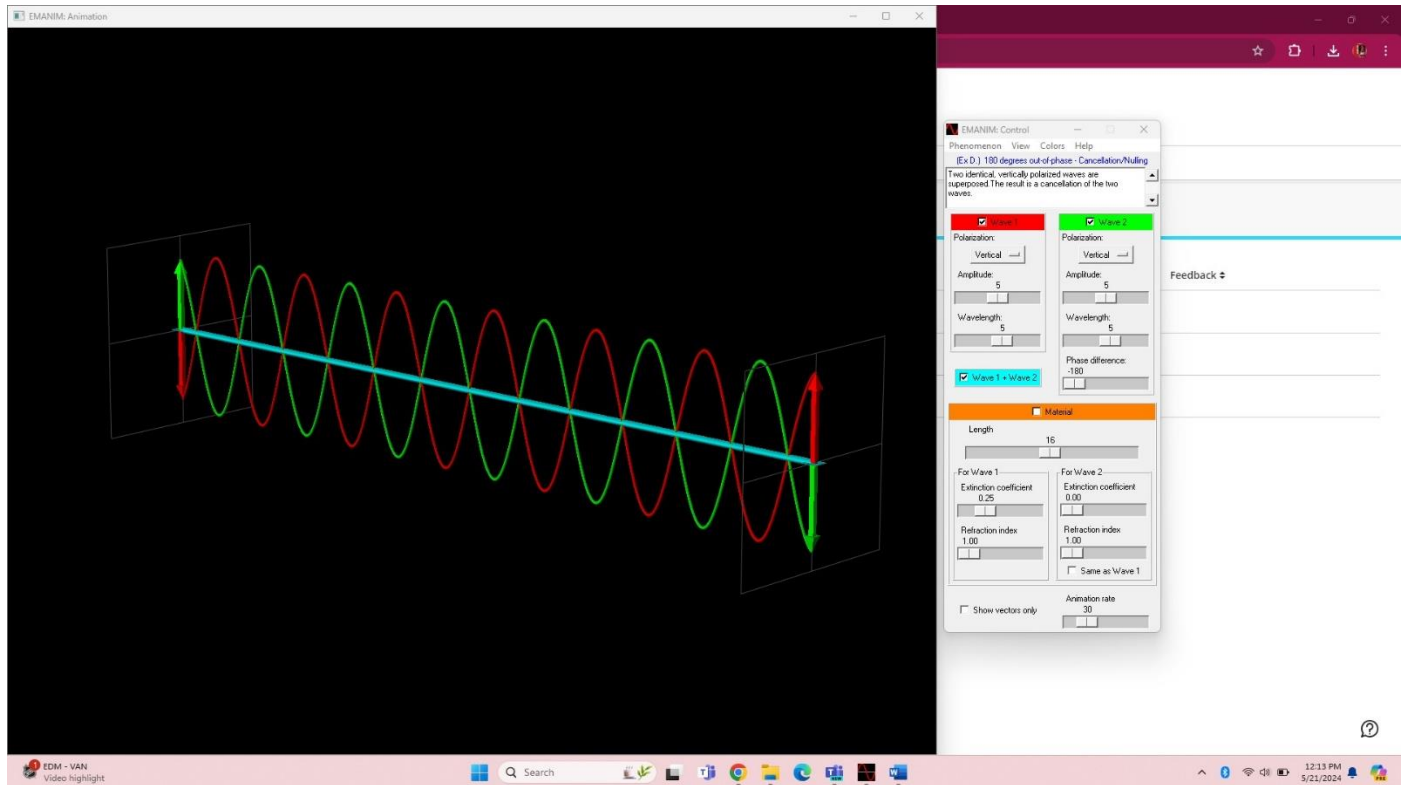
Question: What do we call such case, when we receive a deteriorated signal? Explain your answer and provide values of the phase difference that yield this case.

A destructive multipath is what we term such a circumstance. A reduction in signal intensity occurs when numerous RF signal route that are out of phase with the primary wave arrive at the receiver at the same time. The blue line in the illustration represents the resulting wave, and it is weaker than the other two waves. It happens when the wave's phase discrepancies range from 121 to 179 degrees.

6. Click on Exercise D.

Two identical, vertically polarized waves are superposed. The result is a cancellation of the two waves.

Insert Screen Shot:



Question: What is the value of the received signal? Is it shown in the figure? Explain your answer.

When the two of the same amplitude cancel each other out, the resulting wave also has the same amplitude hence the received signal has zero value. Nulling is another name for it. Sure the resultant wave of the two waves is represented by the blue straight line in the illustration.

Before you proceed to next part, change the polarization of the waves and examine its effect on the final received signal. **Comment on your results.**

The resulting wave is always same when the polarization of the two waves is changed, but the direction oscillation electric field vector of the resulting wave varies based on the type of polarization of the two waves. This implies that when two waves are polarized vertically, the resulting waves will likewise oscillate vertically, and vice versa for the other polarization.

Part 2: Answer the following questions:

1. What is MIMO? Explain briefly.

Answer: The full form of MIMO is Multiple- Input, Multiple- Output. MIMO system takes advantage of multiple paths. MIMO is a wireless radio architecture that can receive or

transmit signal using multiple antennas all at once together. In MIMO device we must make sure that radio channels travel from different signal polarization.

2. What is meant by ISI? Explain briefly.

Answer: **The delay spread time differences can cause bits to overlap with each other and the result will be the corrupted data. This kind of multipath is the common occurrence of destructive multipath.**

3. What is meant by Delay Spread? Explain briefly.

Answer: **Delay Spread means the time differences between multiple paths. This happens because there are several ways for the sent signal to get the receiver including straight visibility and reflections caused by walls, surfaces, and other items.**

4. What are the four possible results of multipath? Explain briefly.

Answer: The four possible results of multipath are:

1. Upfade

Upfade is a increase signal strength. Smaller phase differences between 0 and 120 degrees will cause upfade. When the multiple RF signal paths arrive to the receiver at the same time then they are in phase with primary wave which in result turns to increase in amplitude.

2. Downfade

Downfade means decreased signal strength. When the multiple RF signal paths comes to the receiver at the same time and are out of phase then amplitude decreases. Decreased amplitude of multipath can be considered as destructive multipath.

3. Nulling

Nulling simply means cancellation of signal. When the multiple RF signal paths arrive with the receiver at the same time. Nulling is the complete cancellation of the RF signal.

4. Data corruption

Data corruption is the most common occurrence of destructive multipath. The delay spread time differences which causes bits to overlap with each other and the final result is corrupted data.

5. What is meant by active gain and passive gain?

Answer: **Active gain is caused using an amplifier on the wire which connects the transceiver to the antenna. Active gain devices require the use of an external power source.**

Passive gain focuses the RF signal with the use of antenna then it gets completed. Antennas in passive gain does not require any external power sources.

6. What device is used to display the signal in time domain?

Answer: The device used to display the signal in time domain is Oscilloscope.

7. What device is used to display the signal in frequency domain?

Answer: The device used to display the signal in frequency domain is spectrum analyzer. This tool is used by WLAN engineers.